

Kmos-0009

SUMMARY

This specification defines how the architectural components described in previous KMOS specifications are assembled into complete applications.

VERSION

1.0

STATUS

Draft

PURPOSE

This specification defines how the architectural components described in previous KMOS specifications are assembled into complete applications.

Applications are not the primary architectural unit of KMOS.

Applications are compositions of:

Domains Capabilities Events Knowledge Assets Workflows Governance

Every application is built from the same platform foundation.

ARCHITECTURAL PHILOSOPHY

Applications are experiences.

Capabilities perform work.

Knowledge persists.

Assets provide evidence.

Events coordinate communication.

Applications may be replaced without affecting institutional knowledge.

This separation ensures long-term architectural stability.

APPLICATION COMPOSITION

Every application consists of five layers.

Presentation Layer Application Services Workflow Orchestration Capabilities Platform Foundation

The Platform Foundation includes:

Knowledge Graph Asset Registry Event Bus Workflow Engine Capability Registry Governance Services Identity Platform Services

No application bypasses the platform foundation.

REFERENCE APPLICATIONS

Knowledge Studio Purpose: Explore institutional knowledge. Primary Capabilities: Knowledge Search; Relationship Navigation; Vocabulary Explorer; Timeline Explorer; Research Assistant; Ontology Management. Primary Users: Researchers; Editors; Teachers; Students; Archivists.

Media Studio Purpose: Create and transform media. Primary Capabilities: Media Import; Speech Recognition; Editing; Storyboard Generation; Rendering; Voice Production; Subtitle Generation; Publishing Preparation. Primary Users: Media Teams; Content Producers; Editors.

Preservation Platform Purpose: Digitize, preserve, validate, and enrich institutional knowledge. Primary Capabilities: Document Import; Transcript Processing; Vocabulary Learning; Knowledge Extraction; Evidence Collection; Archive Management; Long-Term Preservation. Primary Users: Archivists; Researchers; Knowledge Curators.

Publishing Studio Purpose: Prepare knowledge for distribution. Primary Capabilities: Metadata Generation; Publishing; SEO; Scheduling; Distribution; Approval; Version Management; Publication Analytics. Primary Users: Publishers; Editors; Marketing Teams.

Research Portal Purpose: Support institutional research. Primary Capabilities: Semantic Search; Citation Discovery; Relationship Analysis; Question Answering; Source Comparison; Historical Navigation. Primary Users: Researchers; Scholars; Students; Editors.

Archive Explorer Purpose: Navigate preserved institutional assets. Primary Capabilities: Asset Discovery; Version Comparison; Lineage Visualization; Evidence Review; Timeline Navigation; Historical Reconstruction. Primary Users: Archivists; Auditors; Researchers.

Learning Platform Purpose: Deliver structured educational experiences. Primary Capabilities: Curriculum Assembly; Lesson Generation; Knowledge Navigation; Assessment; Progress Tracking; Recommendations. Primary Users: Students; Teachers; Institutions.

Mobile Experience Purpose: Provide portable access to institutional knowledge. Examples: MuhammadanWay; Community applications; Offline learning; Personal study; Notifications; Daily learning. The mobile application consumes platform capabilities rather than implementing them.

SHARED PLATFORM SERVICES

Every application shares: Knowledge Graph Asset Registry Workflow Engine Capability Registry Event Bus Identity Governance Search Configuration Observability

Applications never duplicate platform services.

CAPABILITY REUSE

Capabilities are reusable across applications.

Example: Speech Recognition is used by Media Studio, Preservation Platform, Learning Platform, Research Portal, and future applications.

Capabilities remain implementation-independent.

SHARED KNOWLEDGE

Every application contributes to the same institutional knowledge.

Example: Transcript correction performed inside Media Studio Vocabulary updated Knowledge Graph expanded Research Portal immediately benefits Learning Platform improves Publishing Studio receives improved terminology Knowledge grows once and benefits all applications.

SHARED ASSET REGISTRY

Every application registers assets.

Examples: Media Studio creates video assets. Publishing Studio creates publication assets. Research Portal creates research exports. Preservation Platform creates archival packages.

All assets participate in the same lineage framework.

EVENT COLLABORATION

Applications collaborate through events rather than direct integration.

Example: Media Studio publishes TranscriptCorrected Knowledge Platform updates concepts Publishing Studio regenerates metadata Learning Platform refreshes lessons Search updates indexes

Applications remain loosely coupled.

WORKFLOW INTEGRATION

Applications define user experiences. Workflow Engine coordinates execution. Capabilities perform work.

Applications never orchestrate capabilities directly.

This preserves architectural consistency.

HUMAN COLLABORATION

Applications provide user interfaces for: Knowledge Review Editorial Approval Media Production Research Publishing Administration Governance

The platform separates user experience from business execution.

AI COLLABORATION

Applications consume AI capabilities. Applications never embed AI logic directly. AI workers are discovered through the Capability Registry. Applications remain independent of specific AI technologies.

MULTI-TENANCY

Applications may serve: Organizations Departments Projects Communities Educational Institutions Archives Publishers

Knowledge remains logically separated while sharing platform infrastructure.

EXTENSIBILITY

New applications require: Presentation Layer Workflow Definitions Capability Composition User Experience

No changes to platform architecture should be necessary. Applications evolve independently.

REFERENCE SOLUTION EXAMPLE

Lecture enters KMOS Preservation Platform imports media Media Studio creates transcript Knowledge Studio expands concepts Research Portal identifies relationships Publishing Studio produces articles Learning Platform creates lessons Mobile App delivers learning

One institutional workflow powers multiple user experiences.

DEPLOYMENT INDEPENDENCE

Applications may be deployed: Individually Together Cloud Local Hybrid Edge Offline

Platform architecture remains unchanged.

SOLUTION PATTERNS

Recommended application patterns include: Studio Applications Explorer Applications Administrative Applications Automation Applications Research Applications Publishing Applications Learning Applications Public Experiences API Products Connector Applications

Each follows identical architectural principles.

ARCHITECTURAL CONSEQUENCES

Applications become lightweight. Capabilities become reusable. Knowledge becomes centralized. Governance becomes consistent. Development accelerates through composition rather than duplication. Future applications inherit decades of accumulated institutional knowledge.

PLATFORM ECOSYSTEM

KMOS is an ecosystem rather than a product. Applications are members of that ecosystem. Some applications produce knowledge. Some enrich knowledge. Some preserve knowledge. Some publish knowledge. Some teach knowledge. All contribute to institutional memory.

DESIGN PRINCIPLES

Every application SHALL: Consume platform services. Reuse existing capabilities. Publish events. Register assets. Participate in governance. Use institutional knowledge. Avoid business logic duplication. Remain replaceable. Respect domain boundaries. Contribute to institutional memory.

EXAMPLE APPLICATION PORTFOLIO

Core Platform Knowledge Graph Asset Registry Workflow Engine Capability Registry Knowledge Studio Media Studio Publishing Studio Research Portal Archive Explorer Learning Platform Mobile Applications Public APIs Future Applications

The platform expands horizontally by adding applications while preserving a stable architectural core.

STRATEGIC ROADMAP

The first generation of KMOS applications should focus on proving the architecture rather than maximizing feature count.

Suggested order: 1. Knowledge Studio 2. Media Studio 3. Preservation Platform 4. Publishing Studio 5. Research Portal 6. Learning Platform 7. Mobile Experience 8. Public API Platform

Each application validates a different aspect of the KMOS architecture while reusing the same foundational services.

NEXT SPECIFICATION

KMOS-0010 defines the Technical Reference Architecture & Deployment Model.

Previous specifications described the logical architecture of KMOS. KMOS-0010 maps those concepts onto concrete technical building blocks, including: Infrastructure Service topology Deployment patterns Storage architecture Event infrastructure API gateways Observability Security Scalability Disaster recovery Reference deployments

This specification completes the foundational architecture of the Knowledge & Media Operating System by connecting the conceptual design to its implementation architecture.